

Academic Excellence - Personal Best - Giving Back to the Community

Semester 1 Report 2021

Daniel DUNCAN

Year 12

List of Subjects

English Advanced

Biology

Mathematics Advanced

Mathematics Extension 1

Physics

Science Extension

Software Design and Development



Year Adviser: Chau Truong



Principal: Kathryn O'Sullivan



2021 Semester 1 Report - Year 12

Level of Achievement

Achievement Achievement Description

Outstanding	The student has an extensive knowledge and understanding of the content and can readily apply this knowledge. In addition, the student has achieved very high levels of competence in the processes and skills and can apply these skills to new situations.
High	The student has a thorough knowledge and understanding of the content and a high level of competence in the processes and skills and can apply these skills to most situations.
Sound	The student has a sound knowledge and understanding of the main areas of content and has achieved an adequate level of competence in the processes and skills.
Basic	The student has a basic knowledge and understanding of the main areas of content and has achieved a limited level of competence in the processes and skills.
Limited	The student has an elementary knowledge and understanding in a few areas of content and has achieved a very limited level of competence in some of the processes and skills.

NBSC Manly Campus can provide written information, following parent application, showing your child's achievement compared to his or her peer group at our school. This information shows the number of students in each of the five achievement levels based on the "Overall Level of Achievement" for each subject.

Additional Programs and Activities

Duke of Edinburgh

Attendance (Semester 1)

Whole Days Absent	1	Explained Whole Days Absent	1
Unexplained Whole Days Absent	0		



2021 Semester 1 Report - Year 12

Subject & Profile Description		Level Of Achievement			
		Consistently	Often	Sometimes	Rarely
English Advanced	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Biology	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Mathematics Advanced	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Mathematics Extension 1	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			

Subject & Profile Description		Level Of Achievement			
		Consistently	Often	Sometimes	Rarely
Physics	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Science Extension	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Software Design and Development	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			



2021 Semester 1 Report - Year 12

English Advanced

Name: Daniel DUNCAN
Class: 12ENAD2
Teacher: Lisa-Maree Newton

Comment on Progress

Daniel is a thoughtful student who applies himself diligently to class activities in order to extend his knowledge of a variety of texts. He consistently demonstrates a mature approach to his studies and proactively seeks and applies feedback in the drafting process. His vlog assessment demonstrated an excellent range of skills in both textual and multimodal forms, reflecting deep knowledge and understanding. Daniel is encouraged to practise writing in timed conditions, with a focus on integrating specific links to unseen questions and experimenting with more complex sentence structures. He is to be commended for his hard work this semester.

Learning Outcomes

Outcomes	O	H	S	B	L
A student independently responds to, composes and evaluates a range of complex texts for understanding, interpretation, critical analysis, imaginative expression and pleasure.	✓				
A student uses, evaluates and justifies processes, skills and knowledge required to effectively respond to and compose texts in different modes, media and technologies.	✓				
A student critically analyses and uses language forms, features and structures of texts justifying appropriateness for specific purposes, audiences and contexts and evaluates their effects on meaning.	✓				
A student thinks imaginatively, creatively, interpretively, critically and discerningly to respond to, evaluate and compose texts that synthesise complex information, ideas and arguments.	✓				
A student investigates and evaluates the relationships between texts.		✓			
A student evaluates the diverse ways texts can represent personal and public worlds and recognises how they are valued.		✓			

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 1 Report - Year 12

Biology

Name: Daniel DUNCAN

Class: 12BIOA6

Teacher: Vicki Maggs

Comment on Progress

Daniel is a very capable and diligent student whose ability and determination to succeed are reflected in his excellent results. He is a highly active participant in class discussions, offering insightful and interesting comments and he is a valued member of his team in practical sessions. Daniel could improve even further with the completion and thorough revision of summaries of all syllabus points as well as continued practice of HSC style questions as these would aid in his understanding of the literacy requirements needed to gain consistently outstanding grades in examinations.

Learning Outcomes

Outcomes	O	H	S	B	L
Selects and processes appropriate qualitative and quantitative data and information using a range of appropriate media.	✓				
Analyses and evaluates primary and secondary data and information.	✓				
Solves scientific problems using primary and secondary data, critical thinking skills and scientific processes.	✓				
Communicates scientific understanding using suitable language and terminology for a specific audience or purpose.		✓			
Explains the structures of DNA and analyses the mechanisms of inheritance and how processes of reproduction ensure continuity of species.	✓				
Explains natural genetic change and the use of genetic technologies to induce genetic change.		✓			

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 1 Report - Year 12

Mathematics Advanced

Name: Daniel DUNCAN

Class: 12MAA3

Teacher: Josinta Chandra

Comment on Progress

Daniel is a motivated student who approaches his work in a diligent manner. His Semester 1 results indicate an excellent ability to apply skills and techniques to solve complex problems. Daniel always displays a mature and positive attitude towards this subject and his fellow class members.

Learning Outcomes

Outcomes	O	H	S	B	L
Applies calculus techniques to model and solve problems.	✓				
Applies the concepts and techniques of periodic functions in the solution of problems involving trigonometric graphs.	✓				
Applies appropriate differentiation methods to solve problems.	✓				
Applies the concepts and techniques of indefinite and definite integrals in the solution of problems.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 1 Report - Year 12

Mathematics Extension 1

Name: Daniel DUNCAN

Class: 12MAX1B8

Teacher: Philippa Blamires

Comment on Progress

Daniel is a hard working, dedicated student who should be congratulated for his efforts this semester. He has demonstrated a high degree of mastery of the content studied. To improve further, Daniel is encouraged to seek out harder problems in all areas to enhance his ability to answer challenging and multifaceted examination questions.

Learning Outcomes

Outcomes	O	H	S	B	L
Applies techniques involving proof or calculus to model and solve problems.		✓			
Applies concepts and techniques involving vectors to solve problems.		✓			
Applies advanced concepts and techniques in simplifying expressions involving compound angles and solving trigonometric equations.	✓				
Uses calculus in the solution of applied problems, including differential equations and volumes of solids of revolution.		✓			

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 1 Report - Year 12

Physics

Name: Daniel DUNCAN

Class: 12PHYSD1

Teacher: Jamie Wan

Comment on Progress

Daniel is an extremely capable student who is to be congratulated on his improvement and achievements this semester. Daniel conducts exemplary secondary research and applies knowledge of electric and magnetic field interactions on charged particles and currents, producing reports of outstanding quality. He demonstrates extensive knowledge and understanding of electric and magnetic field interactions and can solve problems evaluating their effect on charged particles and currents. Daniel has made enormous progress in interpreting and addressing the specific demands of each question. A methodology of analysing the given stimulus and structuring and planning responses accordingly, will help him to demonstrate more extensive syllabus knowledge. Daniel should experience further success if he continues to set, plan and determinedly work towards his goals with continued motivation and effort.

Learning Outcomes

Outcomes	O	H	S	B	L
Analyses primary data to qualitatively and quantitatively describe the motion of projectiles.		✓			
Explains and analyses the electric and magnetic interactions due to charged particles and currents and evaluates their effect both qualitatively and quantitatively.	✓				
Conducts investigations to collect valid and reliable secondary data and effectively communicates scientific understanding using suitable language and terminology.	✓				
Analyses and evaluates quantitative primary data represented in graphical and tabular forms to draw valid conclusions.		✓			
Describes and explains circular motion, motion of particles in gravitational fields as well as electromagnetic interactions as they apply to technological applications.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 1 Report - Year 12

Science Extension

Name: Daniel DUNCAN

Class: 12SCXTA9

Teacher: Anna Molnar

Comment on Progress

Daniel is a capable and hard-working student who is making excellent progress in Science Extension this semester. He has demonstrated that he can effectively source and evaluate relevant scientific research in order to inform the planning of his own research project, as well as take initiative and solve problems as they arise. With sustained diligence in class and at home, Daniel should continue to have success in this course.

Learning Outcomes

Outcomes	O	H	S	B	L
Analyses and assesses the reliability and validity of secondary resources associated with the scientific research project.	✓				
Interrogates relevant and valid peer-reviewed scientific research to develop a scientific research question and hypothesis.	✓				
Develops a rationale to justify chosen scientific research and creates a detailed plan to investigate the scientific hypothesis.	✓				
Critically analyses valid peer-reviewed scientific research to assess relevance, value and rigor.	✓				
Refines and applies the Working Scientifically processes in relation to scientific research.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 1 Report - Year 12

Software Design and Development

Name: Daniel DUNCAN

Class: 12SDDA5

Teacher: Branko Goykovic

Comment on Progress

Daniel is an exceptional student with an outstanding approach to learning in Software Design and Development. His recent results clearly indicate his depth of understanding, application to study and ability to link practical and theoretical concepts. By revising and enhancing examination techniques, Daniel will further improve on these already impressive results.

Learning Outcomes

Outcomes	O	H	S	B	L
Describes various methods used to construct software solutions and how a computer system stores data.	✓				
Describes the development of different language types and their relationship to emerging technologies.	✓				
Identifies and evaluates legal, social and ethical issues in a range of contexts.	✓				
Identifies, applies and evaluates various modular approaches to software problems.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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